

A = matrix

$$A_{n,m} + B_{n,m} = C_{n,m}$$

خاصية الجمع

ex:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 2 & 5 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 9 & 11 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} \alpha & 2\alpha \\ 3\alpha & 4\alpha \end{pmatrix}$$

خاصية الموزن X
مصفوفة

$$A_{n,m} \cdot B_{m,e} = C_{n,e}$$

$$A_{2,3} B_{3,2} = C_{2,2}$$

$$A_{2,3} B_{3,5} = C_{2,5}$$

$$B_{2,3} C_{2,3} = \text{not exist}$$

$$A_{2,2} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B_{2,3} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 5 \end{pmatrix}$$

find $A \cdot B$ and $B \cdot A$

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 5 \end{pmatrix}$$

$$A_{2,2} \cdot B_{2,3} = C_{2,3}$$

خواص المحدارات

$$\begin{vmatrix} a & b & c \\ a & b & c \\ 1 & 2 & 3 \end{vmatrix} = 0$$

$$\begin{vmatrix} a & a & 1 \\ b & b & 2 \\ c & c & 3 \end{vmatrix} = 0$$

لأن المحدد إذا كان الصف الأول = الصف الثاني
أو العمود الأول يساوي العمود الثاني = المحدد = صف

$$\begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{vmatrix} \quad \text{أو} \quad \begin{vmatrix} a & 1 & 0 \\ b & 2 & 0 \\ c & 3 & 0 \end{vmatrix}$$

∴ كان في المحدد عمود أو صف = أ صفار

يكون فيه المحدد = صف

$$C_{2.3} = \begin{pmatrix} 1.1 + 2.2 & 1.2 + 2.1 & 2.5 \\ 3.1 + 4.2 & 3.2 + 4.1 & 4.5 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 4 & 10 \\ 11 & 10 & 20 \end{pmatrix}$$

$$B_{2.3} \times A_{2.2} = \text{not Assist. } \propto$$

Determinant المحددات

$$1) \Delta = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= (1.4) - (2.3)$$

$$= -2$$

$$\Delta = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix}$$

$$= (1.2) - 0 = 2$$

المحدد 3.3

$$A = \begin{vmatrix} - & + & - \\ 4 & 5 & 6 \\ + & - & + \\ 1 & 2 & 0 \end{vmatrix}$$

Find the value of A

1st

$$1 \begin{vmatrix} 5 & 6 \\ 2 & 0 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 1 & 0 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 1 & 2 \end{vmatrix}$$

$$1(0 - 2 \cdot 6) - 2(0 - 6) + 3(4 \cdot 2 - 5 \cdot 1)$$

$$-12 + 12 + 9 = 9$$

1^{ed}

$$-2 \begin{vmatrix} 4 & 6 \\ 1 & 0 \end{vmatrix} + 5 \begin{vmatrix} 1 & 3 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix}$$

$12 - 15 + 12$

$$-2(-6) + 5(-3) - 2(6 - 12)$$

$$= 12 - 15 + 12 = 9$$

Cramer method for Solving System if linear Equation

Find the Solution the Equation

$$x + y + z = 3$$

$$x - y - z = -1$$

$$x + 2y + 3z = 5$$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 & -1 \\ 2 & 3 & -1 \end{vmatrix} - \begin{vmatrix} 1 & -1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix}$$

$$= \cancel{1(-4-2)} = (-3+2) - (3+1) + (2+1)$$

$$= \cancel{7} (-3+2) - (3+1) + (2+1) \\ -1-4+3 = -2$$

$$\begin{array}{c|ccc}
 & 1 & 2 & 3 \\
 \hline
 \text{عادل مشترك} & 2 & 4 & 6 \\
 2 & & & \\
 \hline
 \text{عادل مشترك} & 5 & 15 & 25 \\
 5 & & & \\
 \hline
 \end{array}
 = 25$$

$$\begin{array}{c|ccc}
 & 1 & 2 & 3 \\
 \hline
 & 1 & 3 & 5 \\
 \hline
 \end{array}
 = 10 \times 0$$

$$= 0$$

$$\Delta = -2$$

$$\Delta X = \begin{vmatrix} 3 & 1 & 1 \\ -1 & -1 & -1 \\ 5 & 2 & 3 \end{vmatrix}$$

$$\Delta X = 3 \begin{vmatrix} -1 & -1 \\ 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} -1 & -1 \\ 5 & 3 \end{vmatrix} + 1 \begin{vmatrix} -1 & -1 \\ 5 & 2 \end{vmatrix}$$

$$= -3 - 2 + 3 = -2.$$

$$\Delta y = \begin{vmatrix} 3 & 1 \\ -1 & -1 \\ 5 & 3 \end{vmatrix}$$

$$1 \begin{vmatrix} -1 & -1 \\ 5 & 3 \end{vmatrix} - 3 \begin{vmatrix} -1 & -1 \\ 1 & 3 \end{vmatrix} + 1 \begin{vmatrix} -1 & -1 \\ 1 & 5 \end{vmatrix} = -4$$

$$\Delta Z = \begin{vmatrix} 1 & 1 & 3 \\ -1 & -1 \\ 2 & 5 \end{vmatrix}$$

$$1 \begin{vmatrix} -1 & -1 \\ 2 & 5 \end{vmatrix} - 1 \begin{vmatrix} -1 & -1 \\ 1 & 5 \end{vmatrix} + 3 \begin{vmatrix} -1 & -1 \\ 1 & 2 \end{vmatrix} = 12$$

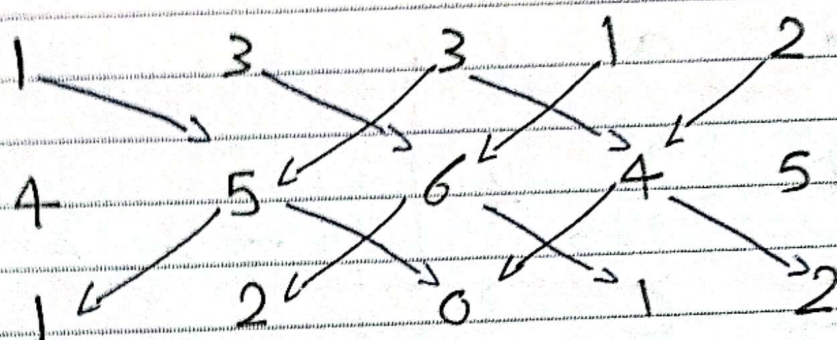
$$X = \frac{\Delta X}{\Delta} = \frac{-2}{-2} = 1$$

$$y = \frac{\Delta y}{\Delta} = \frac{-4}{-2} = 2$$

Given

$$Z = \frac{\Delta Z}{\Delta} = \frac{12}{-2} = 6$$

1	2	3	طريقة سوارس
4	5	6	
1	2	0	



$$\text{القصر الأول} = (1 \cdot 5 \cdot 0 + 3 \cdot 6 \cdot 1 + 3 \cdot 4 \cdot 2) = 36$$

$$\text{القصر الثاني} = -(2 \cdot 4 \cdot 0 + 1 \cdot 6 \cdot 2 + 3 \cdot 5 \cdot 1) = -27$$

$$= 36 - 27 = 9$$

$$\Delta = 9$$